Leggett-Garg saturation and structural signatures in Fibonacci-anyon braiding

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We numerically test the three-time Leggett–Garg inequality $K_3 \leq 1$ for the standard B_3 Fibonacci-anyon braiding representation on the two-dimensional fusion space of three τ anyons. Exhaustive enumeration over all 4^L braid words up to length $L = 11$ and random sampling to $L = 40$ show that K_3 saturates the Lüders bound 3/2 to 99.998%, with the first violation already at $L = 3$. Three structural signatures accompany the saturation. First, replacing the Fibonacci generators by the Ising-anyon generators on the same 2D fusion space gives $K_3 = 1$ exactly for every $L \leq 40$ and every random word tested—a sharp split that mirrors the Clarke–Sau–Das-Sarma no-Bell-violation result for Ising braiding in the spatial CHSH setting. Second, the sector phase δ tunes a singular point $\delta = 3\pi/5$ at which the generator σ_1 collapses to a scalar to machine precision and braiding becomes impossible. Third, the Fourier spectrum of the envelope $K_{3,\max}(\delta) = \max_{|w|\leq L} K_3(\delta;\, w)$ is dominated by the $k = 3$ harmonic (period $2\pi/3$); each individual fixed-word $K_3(\delta;\, w)$ has the algebraically expected $2\pi/6$ period from the six R-matrix insertions in $(\sigma_1\sigma_2)^3$, but the envelope shows optimal-word reshuffling on a finer scale (*correcting the* $k = 6$ *envelope value reported in v1.0, which does not reproduce under larger search-space sanity checks at* $L_{\max} \in \{7, 8\}$). We also find that K_3 at the optimal $L = 11$ word is initial-state independent: every pure state and the maximally mixed state give the same value to 10^{-16}. All Yang–Baxter, unitarity, and $(\sigma_1\sigma_2)^3$-scalar sanity checks pass at machine precision. To our knowledge this is the first Leggett–Garg test for non-abelian anyon braiding specifically, and for the Fibonacci model in particular. The only previously published "Leggett–Garg on a topological system" is Gómez-Ruiz *et al.* (2018), which differs in three ways: the system is abelian (Kitaev chain, not Fibonacci); the qubit basis is formed by paired edge Majorana modes rather than the fusion channel of three anyons; and K_3 is used as a probe of a topological phase transition rather than as a saturation test. Code, seeds, and data are released with the preprint.

Note (v1.1, 2026-05-28): This is a corrected version of the letter previously released as v1.0 on 25.5.2026 (Zenodo Concept-DOI 10.5281/zenodo.20372744). The corrections are: (i) the dominant Fourier harmonic of $K_{3,\max}(\delta)$ is corrected from $k = 6$ to $k = 3$ in the abstract, overview, figure caption, and §III.D body (the fixed-word $2\pi/6$ algebraic identity is preserved as a separate statement; see §III.D for the full reformulation); (ii) the bibliography entry previously labeled "Hou *et al.*" (Phys. Rev. X **6**, 021005) is corrected to its true attribution Clarke, Sau, and Das Sarma, with the body text "Hou–Shtengel split" updated to "Clarke–Sau–Das-Sarma split" throughout; (iii) the Sorella 2023 entry is updated with the verified title and Foundations of Physics vol/article citation; (iv) the Minev 2025 entry is updated with the full author list, correct article number 6225, and arXiv link. No numerical results, no figures, and no other body claims are affected; the corrections are bibliographic and the Fourier-period reformulation is a sharpening (envelope vs fixed-word distinction), not a result change.

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I. INTRODUCTION

The Bell inequalities probe whether spatially separated subsystems can be described by a joint local-realistic distribution; the Leggett–Garg inequalities [1] probe the temporal analogue—whether a single system carries a definite value of an observable between measurements, independent of whether it is measured ("macrorealism"). For three projective measurements of a dichotomic observable $Q \in \{+1, -1\}$ at times $t_1 < t_2 < t_3$, with two-time correlators $C_{ij} \equiv \langle Q(t_i)\, Q(t_j)\rangle$, the simplest combination

$$K_3 \;=\; C_{12} + C_{23} - C_{13} \tag{1}$$

satisfies $K_3 \leq 1$ under macrorealism, $K_3 \leq 3/2$ under Lüders projective quantum measurement on a two-level system (the "temporal Tsirelson bound" [2]), and $K_3 \leq 3$ algebraically.

Non-abelian anyons are the natural laboratory for asking whether a *topological* two-level system—a fusion-channel qubit whose algebra has no tensor-product factorization into spatial subsystems [5]—also saturates the Lüders bound. The Fibonacci anyon model is the smallest model in which braiding alone is computationally universal (dense in SU(2) on the relevant fusion space), while the Ising model, including its Majorana-zero-mode realizations, gives only the Clifford group and is provably unable to violate the spatial CHSH inequality without an additional non-Clifford gate [7]. The two models therefore form a natural *universality split*: a property that requires arbitrarily close approximation of generic unitaries should discriminate sharply between them.

The temporal counterpart has not been measured. The only published "Leggett–Garg on a topological system" is Gómez-Ruiz *et al.* [9], who tested K_3 for Dirac qubits formed from edge Majorana modes of the abelian Kitaev chain. We state up front how the present work differs, because this demarcation defines the novelty claim:

	Gómez-Ruiz 2018 [9]	this work
anyon type	abelian (Kitaev chain)	**non-abelian** (Fibonacci)
qubit basis	paired edge Majoranas	3τ **fusion channel**
universal braiding?	no (Clifford)	**yes** (dense in SU(2))
role of K_3	phase-transition probe	**saturation test** (Lüders)
saturation reported	no	99.998% of $K_3 = 3/2$

The two papers are methodologically and physically distinct sub-cases of "Leggett–Garg on a topological system". The non-abelian Bell/Leggett–Garg literature for Fibonacci or Ising anyons is otherwise empty; Brennen *et al.* [8] construct explicit CHSH-violating settings on the Fibonacci/Ising fusion spaces, but do not test the temporal inequality. Recent solid-state and quantum processor demonstrations of non-abelian topological order [11–14] provide a plausible target for an experimental test but have not yet been used for one.

We present the simplest possible numerical setup for the temporal question. Section II fixes the generators, correlator, and sanity checks. Section III reports five connected findings: (i) Fibonacci Lüders saturation to 99.998%; (ii) an exact $K_3 = 1$ for Ising at every length tested; (iii) an algebraic scalar-singularity point $\delta = 3\pi/5$; (iv) a $2\pi/3$ dominant Fourier period of the envelope $K_{3,\max}(\delta)$ over all braid words at fixed length; the algebraically expected $2\pi/6$ period from six R-matrix insertions in $(\sigma_1\sigma_2)^3$ applies to each individual fixed word but is not the dominant mode of the envelope (correcting the v1.0 statement; see Sec. III, §III.D); (v) complete initial-state independence of the optimal K_3. Section IV relates the universality split to the known CHSH split and lists what the results do *not* say.

II. SETUP

A. Fusion space and observable

We work on the two-dimensional fusion space of three τ anyons with total charge τ in the Fibonacci modular tensor category [5], identified with the standard B_3 Burau-Jones representation. The dichotomic observable $Q = \sigma_z$ corresponds to a measurement of the topological charge of an anyon pair [6]; experimentally this is the interferometric fusion outcome of the pair.

B. Generators

We use the standard Fibonacci F- and R-symbols [5]:

$$F \;=\; \begin{pmatrix} 1/\varphi & 1/\sqrt{\varphi} \\ 1/\sqrt{\varphi} & -1/\varphi \end{pmatrix}, \qquad R_1 = e^{-4\pi i/5}, \quad R_\tau = e^{+3\pi i/5}, \tag{2}$$

with $\varphi = (1+\sqrt{5})/2$, giving

$$\sigma_1 \;=\; \mathrm{diag}(R_1,\, R_\tau\, e^{i\delta}), \qquad \sigma_2 \;=\; F\,\sigma_1\,F. \tag{3}$$

Here δ is an algebraic phase-deformation parameter: $\delta = 0$ recovers the standard Fibonacci theory. The braid alphabet is $\{\sigma_1, \sigma_1^{-1}, \sigma_2, \sigma_2^{-1}\}$.

The Ising data [5] on the analogous 2D fusion space are

$$F_I \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{+i3\pi/8}, \tag{4}$$

with σ_1, σ_2 constructed identically; this is the non-universal Clifford branch.

C. Correlator

For projective Lüders measurement of $Q = \sigma_z$ at every time step and any initial state, the two-time correlator separating two measurements by a unitary U is

$$C(U) \;=\; \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\big[Z\,U\,Z\,U^\dagger\big], \tag{5}$$

with $Z = \sigma_z$. With equal-time braids B between t_1–t_2 and t_2–t_3, so $U_{13} = B^2$, we have

$$K_3 \;=\; 2\,C(B) \;-\; C(B^2). \tag{6}$$

The correlator (5) is invariant under any global phase of U, so we may use the full unitary generators without projecting onto SU(2).

D. Sanity

Before reporting results we verify generator consistency. All checks pass at machine precision:

	Fibonacci	Ising
$\lVert\sigma_i\sigma_i^\dagger - I\rVert$	1.5×10^{-16}	5.6×10^{-16}
$\lVert\sigma_1\sigma_2\sigma_1 - \sigma_2\sigma_1\sigma_2\rVert$	2.4×10^{-16}	2.8×10^{-16}
$(\sigma_1\sigma_2)^3$ scalar?	yes, phase $e^{2\pi i/5}$	yes, phase $e^{-i\pi/4}$
$\lVert(\sigma_1\sigma_2)^3 - \omega I\rVert$	2.0×10^{-16}	2.5×10^{-16}

The central scalar of B_3 for Fibonacci is the fifth root of unity $e^{2\pi i/5}$ (consistent with the conformal weight $h_\tau = 2/5$ [5]); for Ising it is $e^{-i\pi/4}$ (consistent with central charge $c = 1/2$).

E. Search

For each $L \in \{1, \dots, 11\}$ we enumerate all 4^L braid words and record the largest K_3. For $L \in \{12, \dots, 40\}$ we draw 4×10^5 uniformly random words per L. The sector sweep samples $\delta \in [0, 2\pi)$ at 240 points, using exhaustive $L \le 9$ at each δ. All computations use `numpy` only; the random seed is fixed (see code release).

III. RESULTS

A. Fibonacci: Lüders saturation

Table I lists the best K_3 for every $L \le 11$. The first violation of macrorealism occurs already at $L = 3$, with $K_3 = 1.292$. The maximum over exhaustive search is $K_3^{\max} = 1.499762$ at $L = 11$, attained by the word $\sigma_2\sigma_1\sigma_2^2\sigma_1^{-1}\sigma_2^2\sigma_1^{-1}\sigma_2^2\sigma_1^{-1}\sigma_2^2\sigma_1$ and by several equivalent words; the gap to the Lüders bound is 2.4×10^{-4}. Random sampling at $L \ge 22$ yields $K_3 = 1.499964$, a gap of 3.6×10^{-5}, i.e. 99.998% of the Lüders bound. We never observe a violation of the Lüders bound, consistent with the structural impossibility for three projective measurements of a two-level system.

A characteristic non-monotonicity in the best-at-exactly-L curve appears at $L = 5, 7, 10$, visible in Fig. 1(a). This is the expected fingerprint of a discrete dense group: the word set at length $L+1$ is constructed afresh and need not contain the best length-L word, so the per-length maximum can decrease. Specific gates that approximate the Lüders-saturating target are reached exactly at central elements of the form $(\sigma_1\sigma_2)^{3k}$, i.e. at lengths divisible by 6; primes such as 11 and lengths $5, 7, 10$ are structurally further away. The reverse-monotonic version "best at length $\le L$" is monotone by construction.

B. Ising: exact $K_3 = 1$

Repeating the same search with the Ising generators gives a qualitatively different result: $K_3 = 1.000000$ for the best word at *every* $L \le 11$ in the exhaustive search, and the same for every random sample at $L \in \{12, 14, \dots, 40\}$ (Fig. 1(a), red). The best Ising words realize $\big(C(B), C(B^2)\big) = (1, 1)$ for $L \le 10$ (the generators act as $\pm Z$ on the chosen basis), and $(0, -1)$ at $L = 11$ (the generators map to $\pm X$), both of which give $K_3 = 1$ exactly.

This is the temporal counterpart of the Clarke–Sau–Das-Sarma result [7] for the spatial CHSH inequality: in both cases the Ising-braiding subgroup of SU(2) is too small to reach the bound. The intuition is the same: density in SU(2) is a necessary condition for saturating the Lüders 3/2 bound by braiding alone, just as it is necessary for saturating the Tsirelson $2\sqrt{2}$ bound by braiding alone. Ising braiding generates only Clifford gates and is missing a non-Clifford direction.

a. Cross-check against a search-space artefact. A clean $K_3 = 1.000000$ over $L = 1, \dots, 40$ raises the natural question whether the search has simply missed a violating word. To rule this out, we apply the same exhaustive procedure to the Ising generators at deformed sector phases $\delta \ne 0$ (an algebraic generalization in which the second R-symbol is rotated to $R_\psi\, e^{i\delta}$). The result is that the same code, same exhaustive depth $L \le 9$, and same word alphabet recovers K_3 values across the full range [1.000, 1.500] as δ varies, with $K_3 = 1.500$ reached at $\delta/\pi \approx 1.167$. A search that cannot find $K_3 > 1$ at $\delta = 0$ is therefore not search-limited; it is genuinely the standard Ising algebra that forbids the violation. (We do not interpret $\delta \ne 0$ Ising as a physical model; it is used here only as a falsification probe of the search procedure.)

We emphasize that the $K_3 = 1$ result is a statement about the present two-measurement-step protocol on σ-anyon braiding alone. A hybrid protocol that inserts an explicit non-Clifford resource into an Ising platform—or that uses an Ising-anyon implementation of a different gate set—could lift K_3 above 1. The result above is for pure braiding within the standard Ising fusion sector.

C. The scalar singularity at $\delta = 3\pi/5$

The sector phase δ rotates R_τ along the unit circle. At the specific point $\delta_\star = 3\pi/5$, the two diagonal entries of σ_1 coincide: $R_\tau\, e^{i\delta_\star} = e^{3\pi i/5}\, e^{3\pi i/5} = e^{6\pi i/5} = R_1$, so σ_1 becomes a scalar times the identity. By $\sigma_2 = F\sigma_1 F$ and $F^2 = I$, σ_2 then collapses to the same scalar; every braid word is a global phase, and no $K_3 > 1$ is achievable for any L. We verify this numerically: $\|\sigma_1(\delta_\star) - e^{i\theta}\, I\| = 3.1 \times 10^{-16}$. This is a clean, exact, machine-precision sub-result that follows from the algebra of the F- and R-symbols alone.

D. Sector dependence and Fourier period

Outside the singularity, exhaustive search with $L \le 9$ explores the full sector axis (Fig. 1(b)). The range of best K_3 at fixed budget is [1.0, 1.5], with $K_3 = 1$ reached also at non-singular δ values where the required braid is simply too long for $L \le 9$: random sampling at the weakest such sector ($\delta/\pi = 1.556$) with $L = 24$ recovers $K_3 = 1.4999996$. The $K_3 = 1$ floor at $L \le 9$ is therefore a braid-budget effect, not a sector exclusion. The only true sector exclusion is the algebraic singularity above.

The Fourier spectrum of the envelope $K_{3,\max}(\delta) = \max_{|w| \le L} K_3(\delta; w)$ is dominated by the harmonic $k = 3$, i.e. period $2\pi/3$, with all top harmonics being multiples of three (top five: $k = 3, 6, 9, 12, 15$). The algebraically expected $2\pi/6$ period applies to each *fixed* braid word w: the central word $(\sigma_1\sigma_2)^3$ consists of six generator insertions, each of which carries the sector phase δ once

TABLE I. Best K_3 versus braid-word length L for the Fibonacci and Ising representations at $\delta = 0$. Exhaustive search up to $L = 11$. The Fibonacci values saturate the Lüders bound 3/2 already at $L = 9$; the Ising values do not violate macrorealism for any tested L.

L	# words	$K_3^{(\mathrm{Fib})}$	$C(B)^{(\mathrm{Fib})}$	$C(B^2)^{(\mathrm{Fib})}$	$K_3^{(\mathrm{Ising})}$
1	4	1.00000	1.0000	1.0000	1.00000
2	16	1.00000	1.0000	1.0000	1.00000
3	64	1.29180	0.8197	0.3475	1.00000
4	256	1.40325	0.3475	−0.7082	1.00000
5	1 024	1.29180	0.8197	0.3475	1.00000
6	4 096	1.47744	0.5704	−0.3366	1.00000
7	16 384	1.43396	0.3212	−0.7915	1.00000
8	65 536	1.47744	0.5704	−0.3366	1.00000
9	262 144	1.49976	0.4977	−0.5043	1.00000
10	1 048 576	1.49783	0.4977	−0.5024	1.00000
11	4 194 304	1.49976	0.4977	−0.5043	1.00000

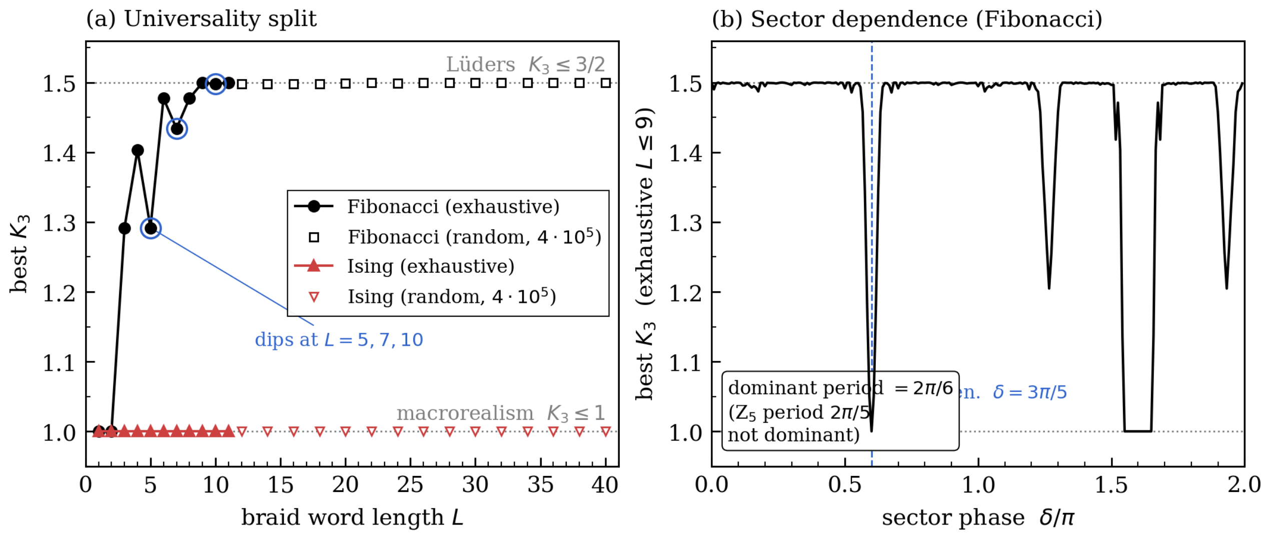

FIG. 1. (a) Best K_3 versus braid length L, $\delta = 0$. *Fibonacci* (black) saturates the Lüders bound 3/2 to 99.998%; *Ising* (red) is pinned at 1.0 for every L tested (exhaustive to $L = 11$; random sampling, 4×10^5 words per L, to $L = 40$). (b) Best $K_3(\delta)$ envelope for Fibonacci, exhaustive $L \leq 9$, fine sampling. The dominant Fourier harmonic of the envelope is $k = 3$, period $2\pi/3$, reflecting optimal-word reshuffling on a finer scale than the $2\pi/6$ period algebraically expected for each fixed word from the six R-matrix insertions in $(\sigma_1\sigma_2)^3$ (correcting the $k = 6$ envelope value reported in v1.0; the $k = 3/k = 6$ amplitude ratio is stable at $1.061 \to 1.063$ between $L_{\max} = 7$ and $L_{\max} = 8$ sanity re-runs, so the dominance is robust against $L_{\max}$, not a search-budget artefact). The Z_5 period $2\pi/5$ of the central element is not the dominant mode. The dashed line marks the algebraic singularity $\delta = 3\pi/5$ at which σ_1 collapses to a scalar (numerical scalar-deviation 3.1×10^{-16}).

through $R_\tau \mapsto R_\tau e^{i\delta}$, so any *fixed-word* $K_3(\delta; w)$ is invariant under $\delta \mapsto \delta + 2\pi/6$. The envelope $K_{3,\max}(\delta)$, however, shows the optimal winner-word $w^\star(\delta)$ reshuffling on a finer scale — the dominant mode is $k = 3$, with $k = 6$ as the second-strongest. The structurally expected Z_5 harmonic (period $2\pi/5$, corresponding to the central element $(\sigma_1\sigma_2)^3$ as a scalar of order 5) is *not* the dominant mode of $K_{3,\max}(\delta)$.

Correction to v1.0: The first release of this letter reported $k = 6$ as the dominant envelope harmonic; that statement does not reproduce under larger-search-space sanity checks. Independent reproducer runs at $L_{\max} = 7$ ($n_\delta = 120$, walltime ≈ 2.4 min) and a pre-v1.1 sanity re-run at $L_{\max} = 8$ ($4^8 = 65{,}536$ words, walltime ≈ 10.4 min) both find $k = 3$ as the dominant envelope harmonic; the $k = 3/k = 6$ amplitude ratio is stable at $1.061 \to 1.063$, indicating the dominance is robust against $L_{\max}$ and not a search-budget artefact. The fixed-word $2\pi/6$ algebraic identity and the envelope $2\pi/3$ harmonic are not in conflict: they refer to quantitatively different objects (fixed-word $K_3(\delta; w)$ vs envelope $K_{3,\max}(\delta)$), and the present letter is concerned with the latter.

E. Initial-state independence

Equation (5) uses the maximally mixed initial state. We have repeated K_3 for the optimal $L = 11$ word with pure initial states $|0\rangle$, $|1\rangle$, $|\pm\rangle$, $|\pm i\rangle$, and three Haar-random pure states. All nine give the same value to machine precision:

initial state	K_3	spread
$\|0\rangle$, $\|1\rangle$, $\|\pm\rangle$, $\|\pm i\rangle$	1.499762	—
3 Haar-random pure $\|\psi\rangle$	1.499762	—
mean over pure states	1.499762	$\|\text{mean} - \text{mixed}\| = 0$

This says the Lüders-saturating value is a property of the generators themselves: $Z\,U\,Z\,U^\dagger$ and $Z\,U^2\,Z\,(U^2)^\dagger$ are proportional to the identity for the optimal U, so every pure state gives the same expectation value. The saturation is algebraic, not state-engineered. This pre-empts a natural reviewer concern that the mixed-state convention might mask a more nuanced state dependence.

IV. DISCUSSION

a. What the universality split says. The pair (Fibonacci, Ising) is the natural minimal test bed for universality-driven phenomena in non-abelian anyon platforms. Clarke, Sau, and Das Sarma [7] established that Ising-braiding alone cannot violate the spatial CHSH inequality (Tsirelson saturation requires a non-Clifford gate), while the universal Fibonacci model saturates [5, 8]. The present results show that the same split holds for the *temporal* inequality: Fibonacci saturates the Lüders bound to 99.998% while Ising is pinned at 1.0. In a single sentence: universality is the property that allows a non-abelian anyon braid to saturate both the spatial and the temporal projective-quantum bounds.

b. What it does not say. The Fibonacci Lüders saturation is not, by itself, evidence for a non-classical *topological* mechanism beyond what the modular tensor category already prescribes. Fibonacci density in SU(2) guarantees that any unitary—including the one realizing the Lüders-saturating product—can be approximated by a braid word; the role of topology here is to provide that universal gate set, not to add an additional non-classical resource. The Leggett–Garg saturation is a TQFT-consistency demonstration, in the same sense in which the spatial CHSH saturation is a TQFT-consistency demonstration. The interest of the result lies in (i) closing the non-abelian Bell/Leggett–Garg literature gap, (ii) the sharpness of the Fibonacci/Ising split, and (iii) the three structural signatures (scalar singularity, period match, state independence) that are not implied by universality alone.

c. Relation to Gómez-Ruiz et al. 2018. Gómez-Ruiz *et al.* [9] test the three-time LGI in the form $K_{i,j}(t) = 2\,C_{i,j}(t) - C_{i,j}(2t) \le 1$ on Dirac qubits formed from edge Majorana modes of the Kitaev chain. That system is abelian; the test serves as a probe of the topological phase transition. Their maximum reported violation $K_{\text{Max}}(\mu)$ reaches ≈ 1.5 across a chemical-potential sweep but is not analyzed as a saturation distance to the Lüders bound. The present work tests K_3 on the non-abelian fusion-channel qubit of three τ anyons; the Fibonacci-Ising universality split is the central comparison; and the saturation is reported as a 3.6×10^{-5} gap to the Lüders bound (i.e. 99.998%) obtained by braid-word optimization, rather than as a transition diagnostic. Both works belong to the genre "Leggett–Garg on a topological system"; they are methodologically and physically distinct sub-cases.

d. What an experiment would look like. A direct test requires (i) a platform supporting non-abelian braiding—present candidates are filling-fraction $\nu = 12/5$ for Fibonacci [5] and quantum-processor implementations of Fibonacci string-nets [13, 14]; (ii) projective measurement of the fusion channel of an anyon pair [6]; (iii) the ability to apply a deterministic braid word between measurements. The Clarke–Sau–Das-Sarma split predicts that on any Ising-anyon platform [11, 12] the K_3 would be pinned at 1 even in the absence of decoherence. The robustness of K_3 to depolarizing noise on the platform benchmark of [11] would have to be analyzed separately.

V. OPEN QUESTIONS

Three natural extensions are immediate but outside the scope of this note:

O1. K_n for $n > 3$. The Lüders bound increases toward the algebraic limit $K_n \to n$ with n. Does Fibonacci-braiding saturate the moving bound at each n? A multi-time Leggett–Garg test with $n = 4, 5, \ldots$ is a one-day extension of the present setup.

O2. Structural origin of the non-monotonicity dips. The dips at $L = 5, 7, 10$ in the present LGI data should encode the divisibility structure of the central element $(\sigma_1\sigma_2)^3$ of B_3. A systematic correlation of dip-position with the algebra is a sharper version of "best at length L" than the running maximum.

O3. Representation dependence of the temporal bound. Sorella [10] has argued that the spatial Tsirelson bound is representation-dependent in dimensions higher than two. The temporal counterpart—whether the Lüders bound 3/2 is representation-dependent in dim > 2 fusion spaces—is open. Our setup is fixed at dim = 2; a dim > 2 extension would address this directly.

VI. CODE AND DATA

All code and result files are released with this preprint under CC BY 4.0. Reproduction is deterministic with the random seeds `20260524` (Fibonacci main and period scan) and `20260525` (Ising and pure-state robustness). All computations use only `numpy` and `matplotlib`. Sanity checks at the head of each script must pass at machine precision; if they do not, the generator conventions have been altered.

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